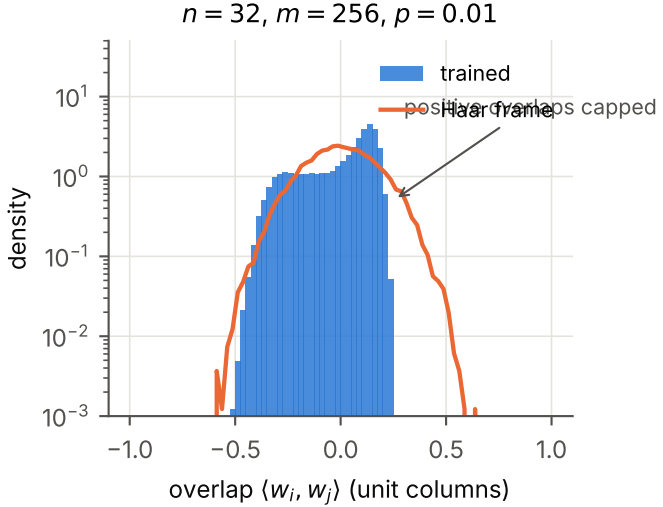
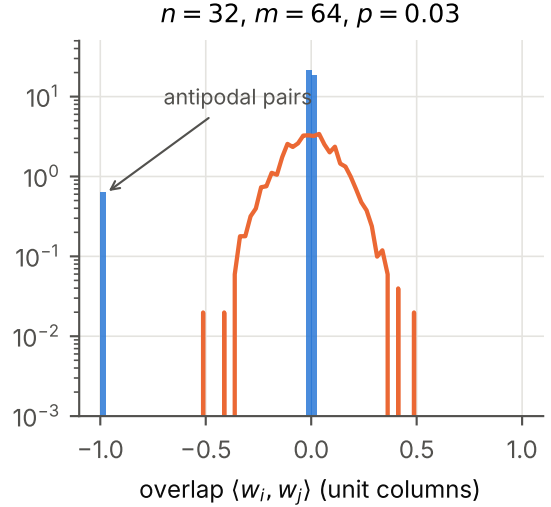
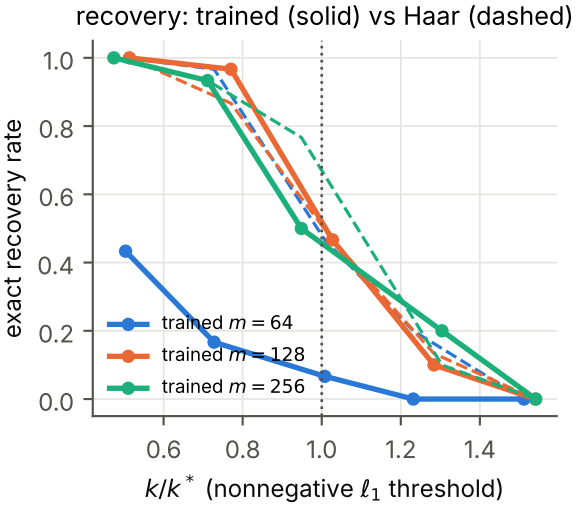

$n = 32$, $m = 256$, $p = 0.01$
trained
positive overlaps capped
Haar frame
density
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
−1.0
−0.5
0.0
0.5
1.0
overlap $\langle w_i, w_j \rangle$ (unit columns)


$n = 32$, $m = 64$, $p = 0.03$
antipodal pairs
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
−1.0
−0.5
0.0
0.5
1.0
overlap $\langle w_i, w_j \rangle$ (unit columns)


recovery: trained (solid) vs Haar (dashed)
exact recovery rate
1.0
0.8
0.6
0.4
0.2
0.0
trained $m = 64$
trained $m = 128$
trained $m = 256$
0.6
0.8
1.0
1.2
1.4
k/k^* (nonnegative ℓ_1 threshold)